

Computational Chemistry – Practical Exam 2023

1.Generation and visualisation of molecular orbitals **CO₂,NH₃,H₂O,N₂:**

Aim:

To generate and visualize molecular orbitals using computational chemistry.

Theory:

A molecular orbital is a mathematical function that describes the location and wave-like behavior of an electron in a molecule.

It is formed by the linear combination of atomic orbitals, which are the orbitals of individual atoms.

Procedure:

1.Build a CO₂, NH₃,H₂O,N₂ atoms in Chemcompute software.

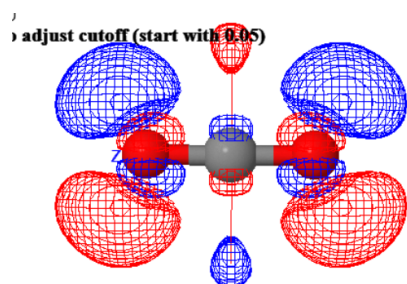
2. Click or choose single point energy calculation.

3. Choose correct basis set in chemcompute.

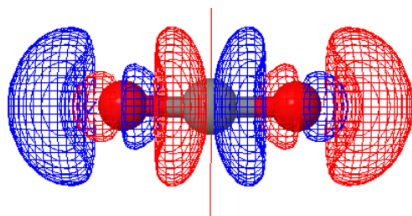
Result:

1. CO₂ :

LUMO



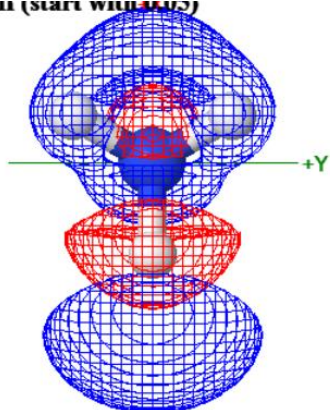
HOMO



2.NH3:

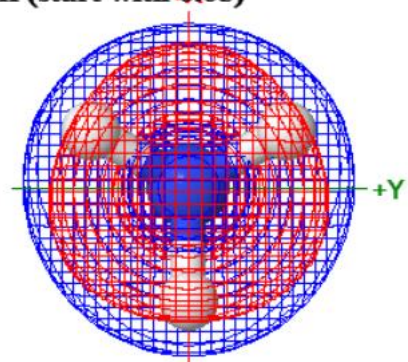
LUMO

non (start with π_{H2S})



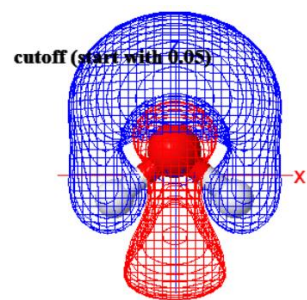
HOMO

non (start with π_{H2S})

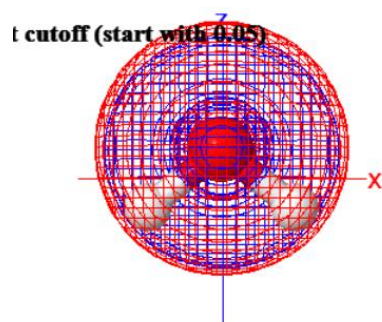


3.H₂O:

LUMO

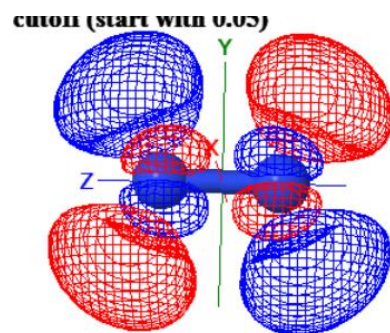


HUMO

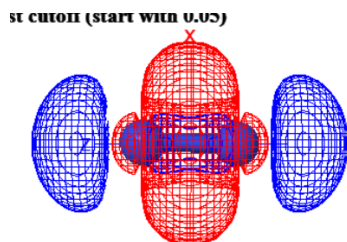


4.N₂:

LUMO



HOMO



2. Energy minimisation of Diatomic Molecules

F₂, H₂, CCl₄ :

Aim:

To do Energy Minimisation of Diatomic Molecules
F₂, H₂, CCl₄.

Theory:

Energy minimisation involves locating the energy minimum of the system simulated under the action of the specified potential model.

We change the geometry of the molecule in such a way that energy gets minimum. This will be the most stable structure, and known as “Optimized geometry”.

Procedure:

1. Build the molecule using Chemcompute server or any other molecular builder.
2. Prepare the input for energy minimization calculation. Choose an appropriate method and basis set.
3. Submit the job.
4. Note optimized energy of the molecule.
5. Report the cartesian optimized geometry of the molecules.

Result:

Molecules	Optimized Energy	Initial Bond Length	Optimized Bond Length
F2	-35.5340847117	1.3400000	1.3400000
H2	-1.0115831104	0.7000000	
CCl4	This is a Polyatomic Molecule.		